

Genetic Algorithm: Order Crossover (OX) on VRP Chromosomes

Parent 1:

[D→1→2→3→D] [D→4→5→D]

Parent 2:

[D→3→1→4→D] [D→2→5→D]



OX Crossover

Child 1:

[D→1→2→3→D] [D→4→5→D] (inherits P1 segment, fixes P2 order)

Child 2:

[D→3→1→4→D] [D→2→5→D] (inherits P2 segment, fixes P1 order)



Repair infeasible routes (capacity check) → Mutation (swap 2 customers) → Next gen